Updated February 4, 2026

Homework problems for AMAT 300 (Intro to Proofs), Spring 2026. Over the course of the semester I'll add problems to this list, with each problem's due date specified. Each problem is worth 2 points.

Solutions will be gradually added (and may be hastily written without proofreading).

Problem 1 (due Weds 2/4): Let $A = \{20a - 26b \mid a, b \in \mathbb{Z}\}$ and $B = \{2c \mid c \in \mathbb{Z}\}$. Prove that $A = B$. (Do it directly, don't use any number theory tricks you might happen to know.)

Problem 2 (due Weds 2/4): Prove that the cardinality of $\{1, \{1\}, \{1, \{1\}\}, \{1, 1\}, \{\{1\}, 1\}\}$ is three. (You need to argue that it has exactly three elements, no more, no less.)

Problem 3 (due Weds 2/4): Let A and B be non-empty finite sets such that $|\mathcal{P}(A \times B)| = |\mathcal{P}(A) \times \mathcal{P}(B)|$. Prove that A and B each have exactly two elements. (Careful not to *assume* they each have exactly two elements, you have to *prove* that they do.)

Problem 4 (due Weds 2/11): Let A and B be sets such that $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ and suppose there exists $b_0 \in B \setminus A$. Prove that $A \subseteq B$.

Problem 5 (due Weds 2/11): Let $\Lambda = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ be the closed disk of radius 1 centered at the origin. For each $\alpha \in \Lambda$, say $\alpha = (x_0, y_0)$, let $X_\alpha = \{(x, y) \in \mathbb{R}^2 \mid (x - x_0)^2 + (y - y_0)^2 = 1\}$ be the closed disk of radius 1 centered at α. Prove (rigorously) that $\bigcup_{\alpha \in \Lambda} X_\alpha = D$, where $D := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 4\}$ is the closed disk of radius 2 centered at the origin.

Problem 6 (due Weds 2/11): With the same setup as the previous problem, prove (rigorously) that $\bigcap_{\alpha \in \Lambda} X_\alpha = \{(0, 0)\}$.

Problem 7 (due Weds 2/18):

Problem 8 (due Weds 2/18):

Problem 9 (due Weds 2/18):
